Model Tuning

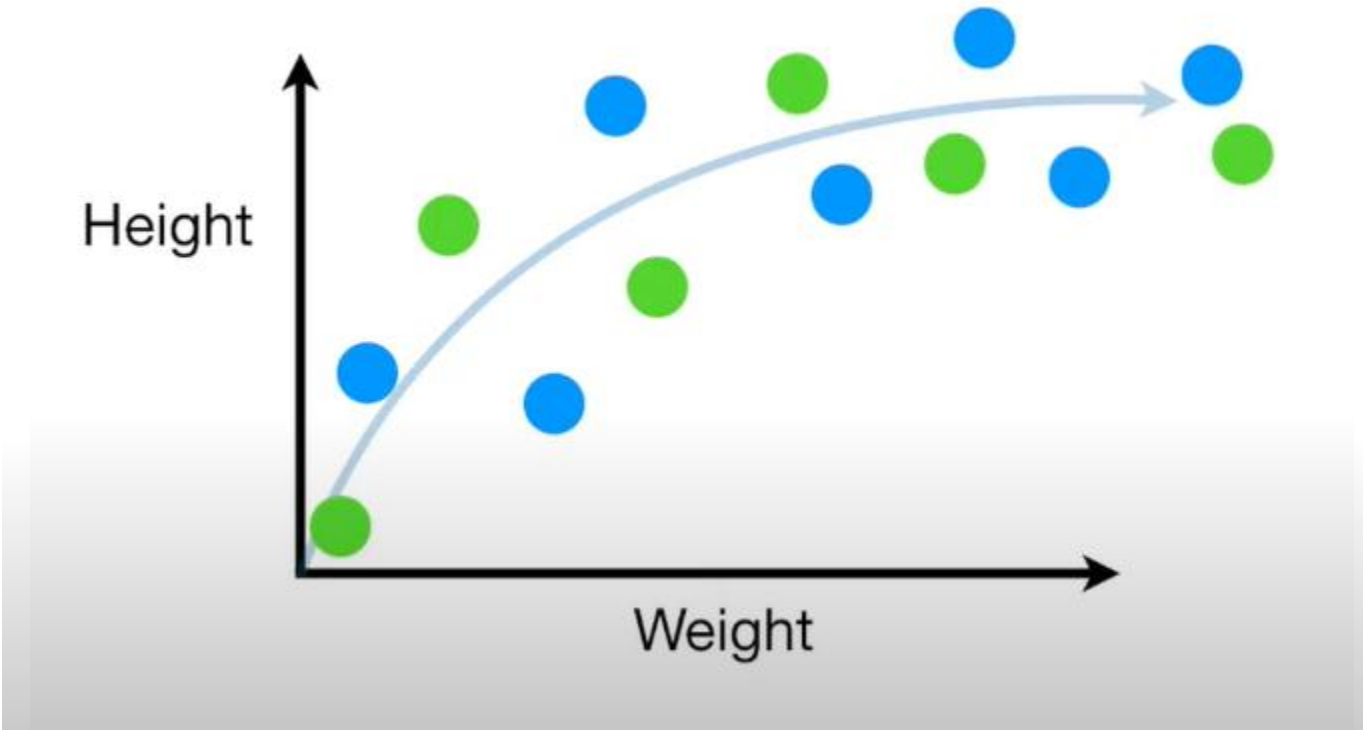
Week 12

Today’s Topics

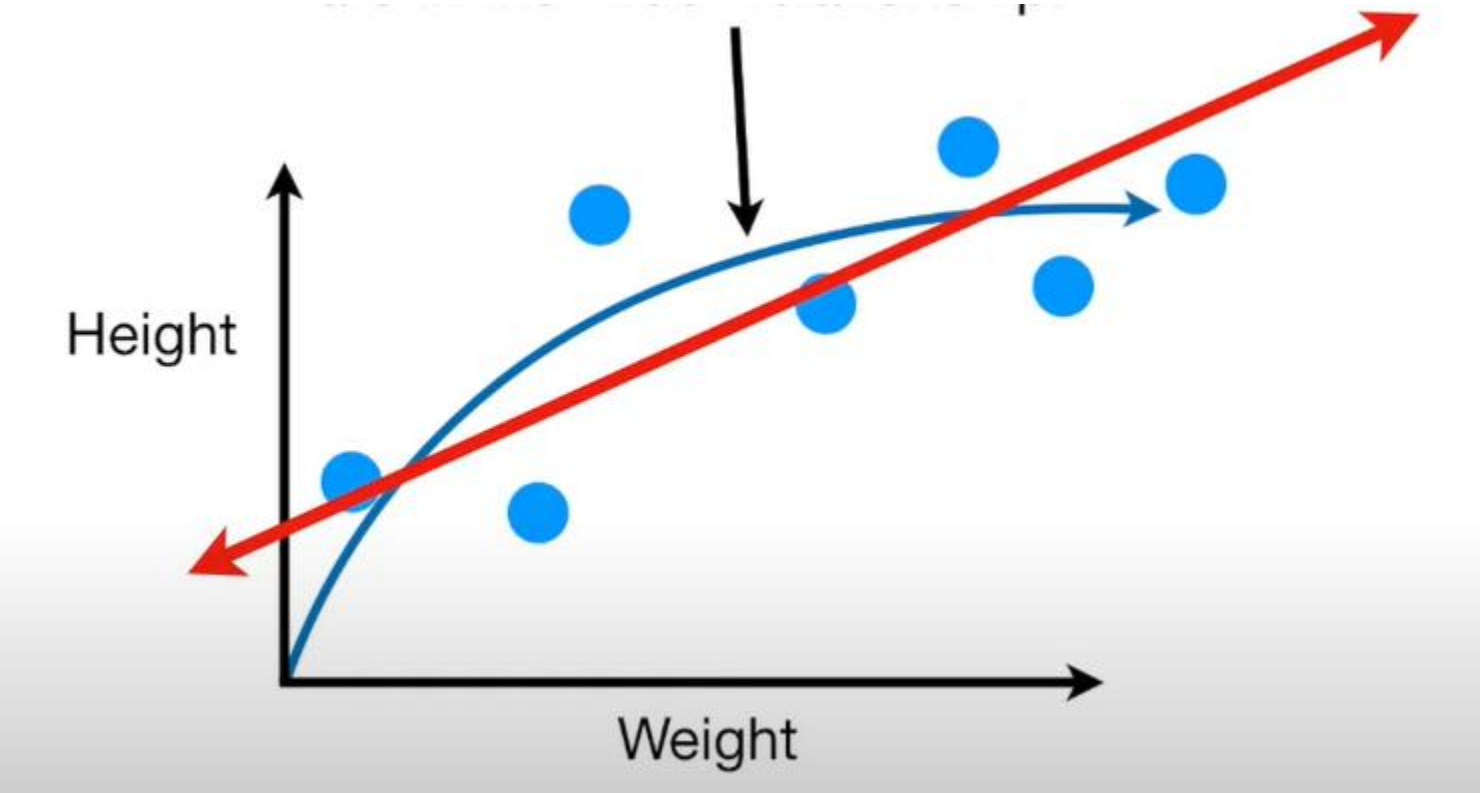
- Model complexity
  - Bias variance trade-off
- Hyper-parameter tunning
  - GridSearchCV
  - RandomSearchCV

Model complexity

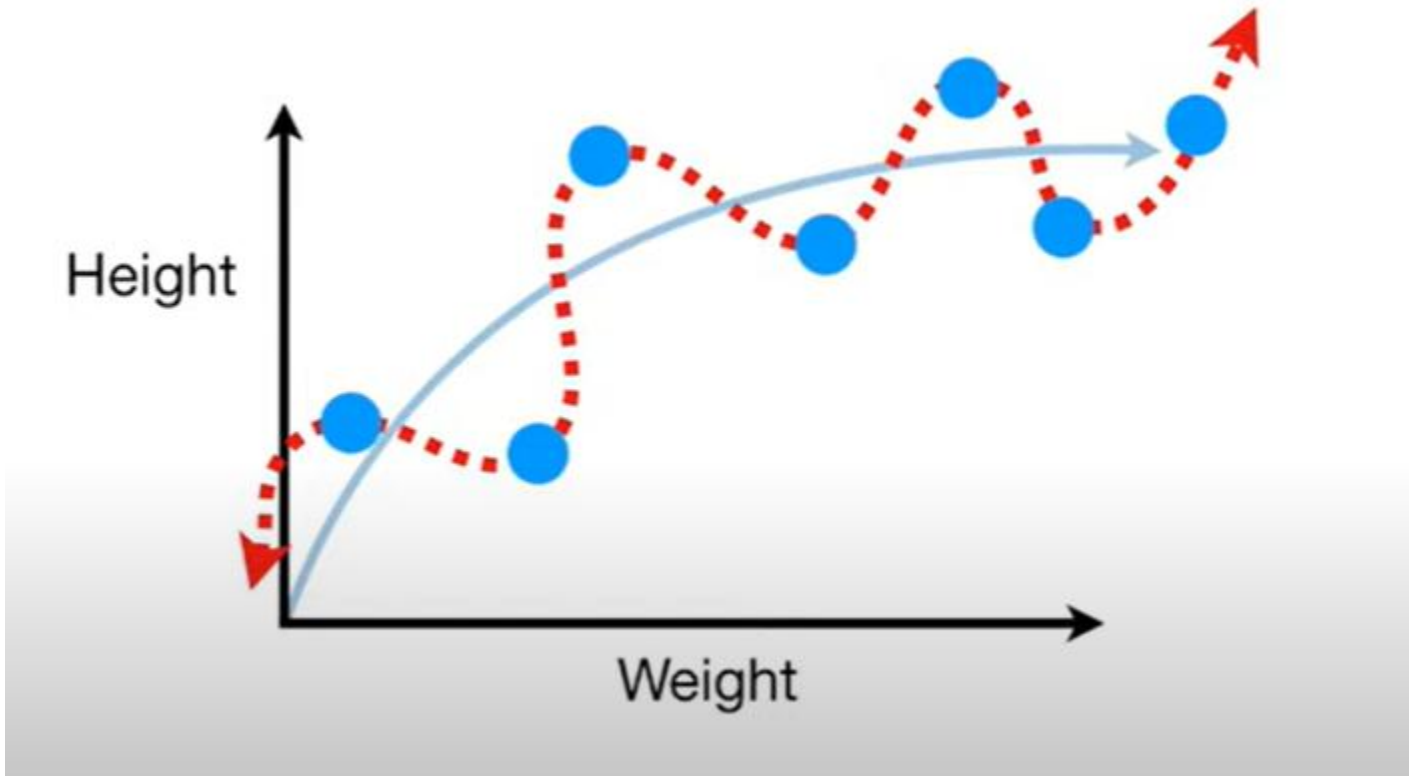
Bias and Variance



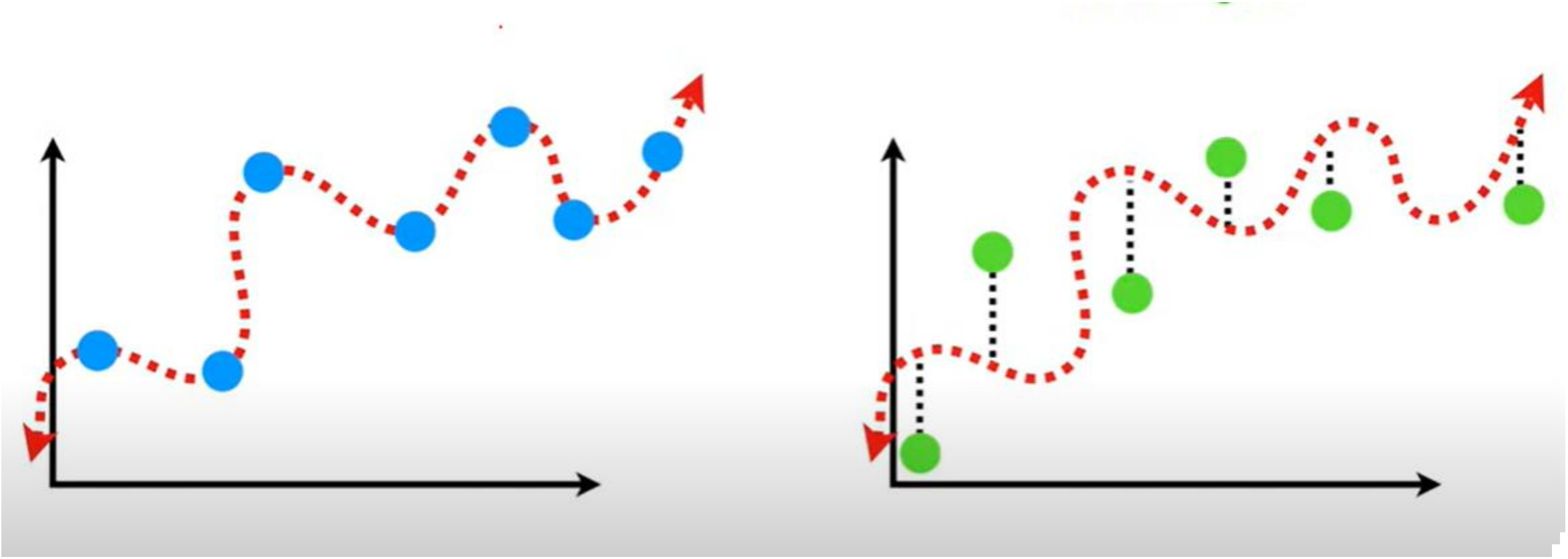
Bias and Variance (Cont....)



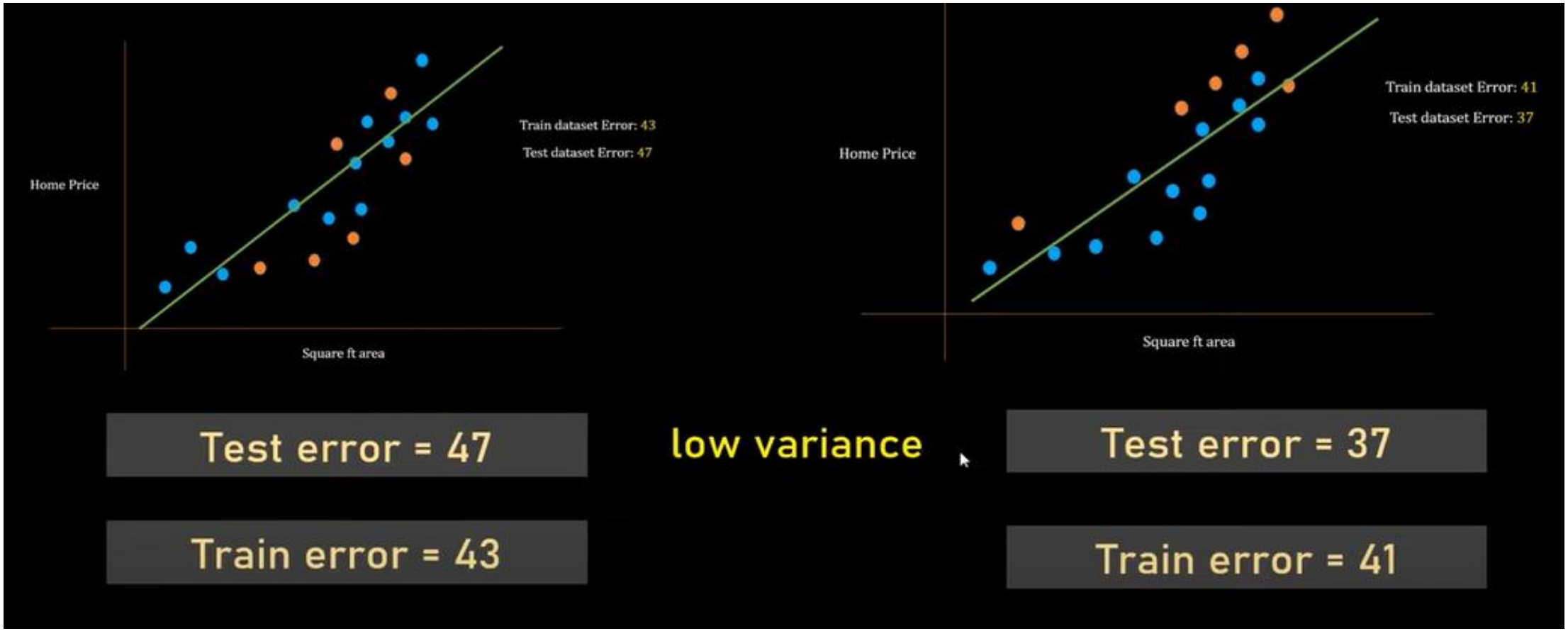
Bias and Variance (Cont....)



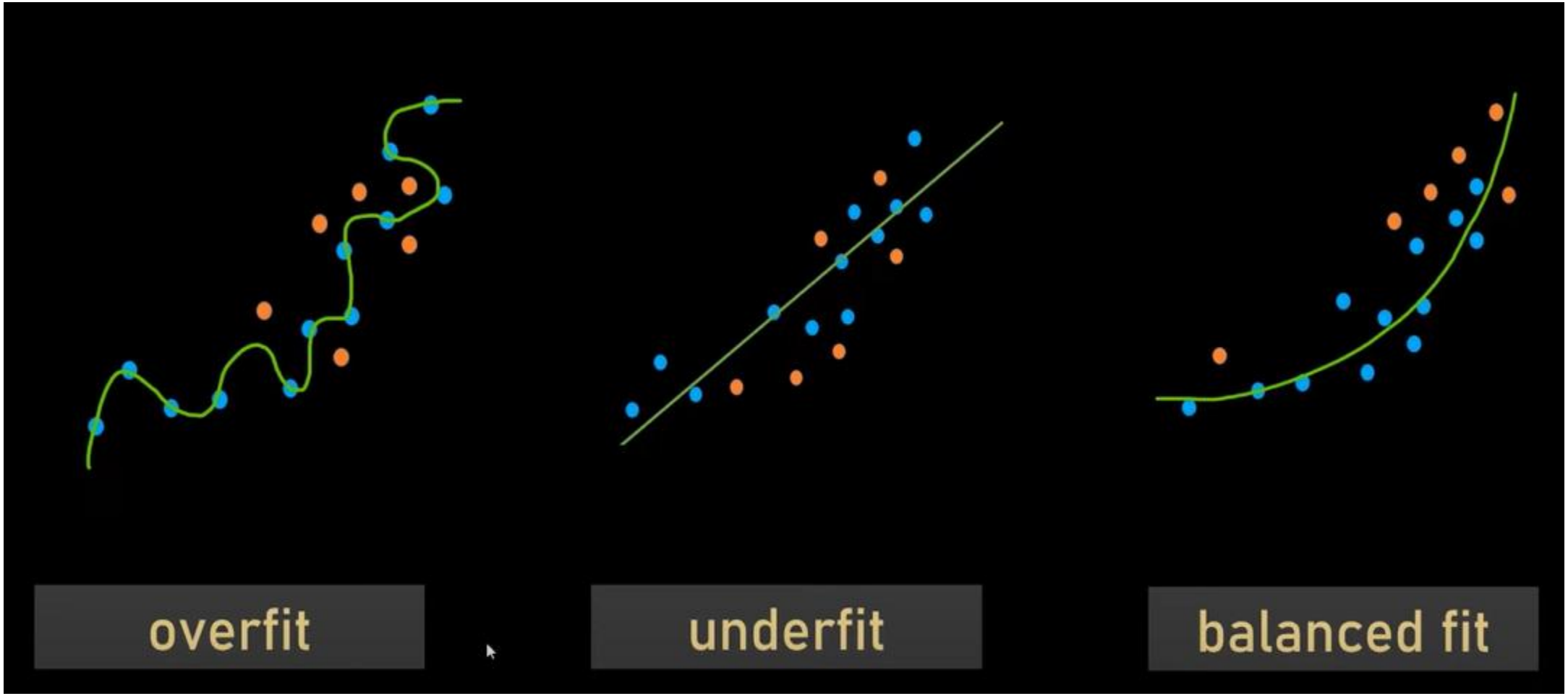
Bias and Variance (Cont....)



Bias and Variance (Cont....)



Bias and Variance (Cont....)



Hyperparameter Tuning

GridSearchCV

RandomSearchCV

Types of Parameters

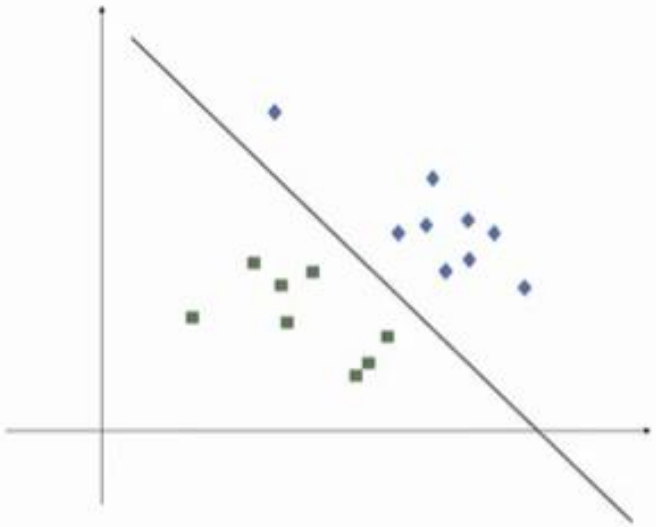
Parameters

Model Parameters

Hyperparameters

Support vector machine draws a hyper plane in n dimensional space such that it maximizes margin between classification groups

2D



3D

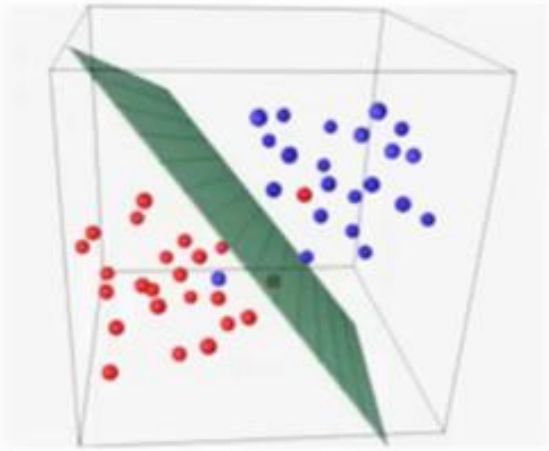
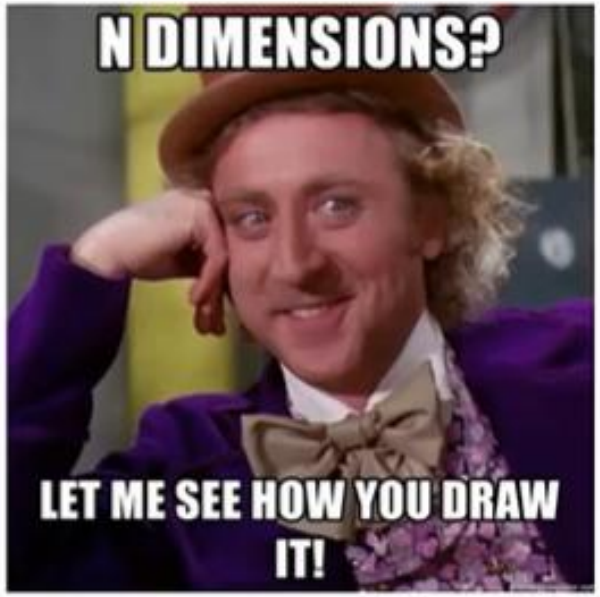
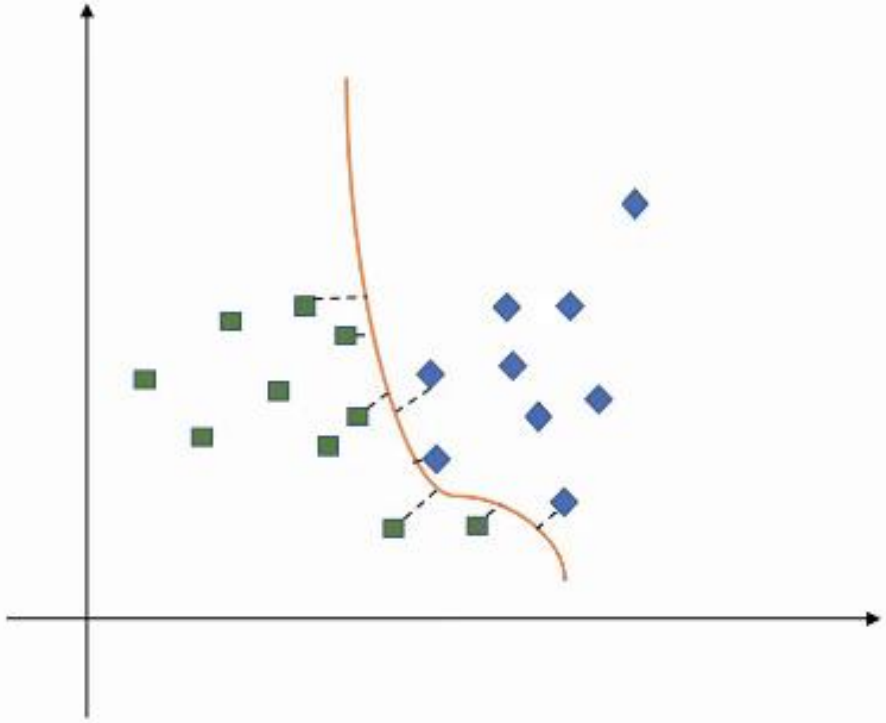


Image Credit: https://appliedmachinelearning.blog

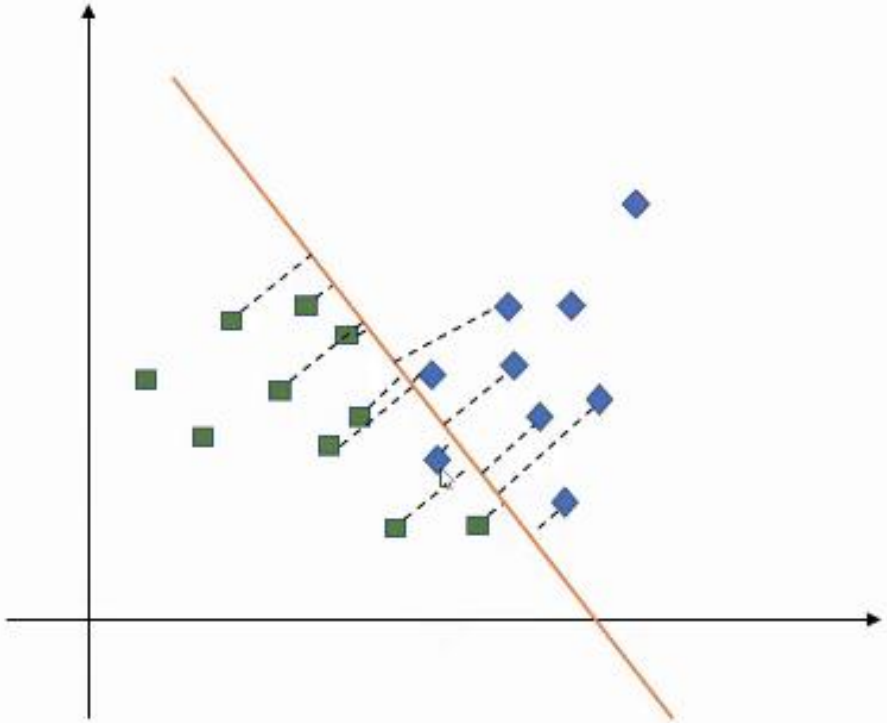
nD



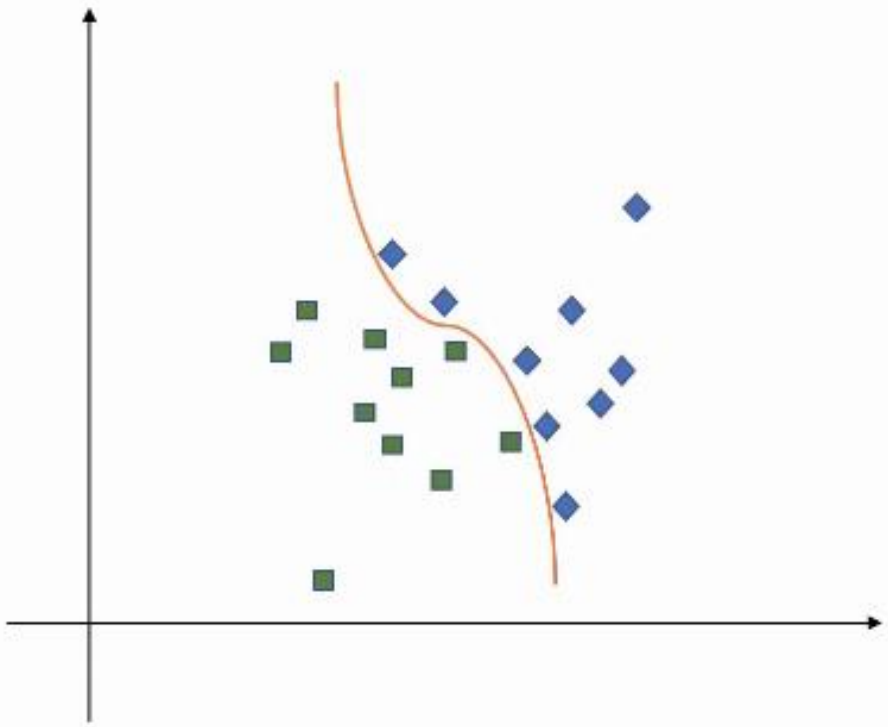
Gamma & Regularization



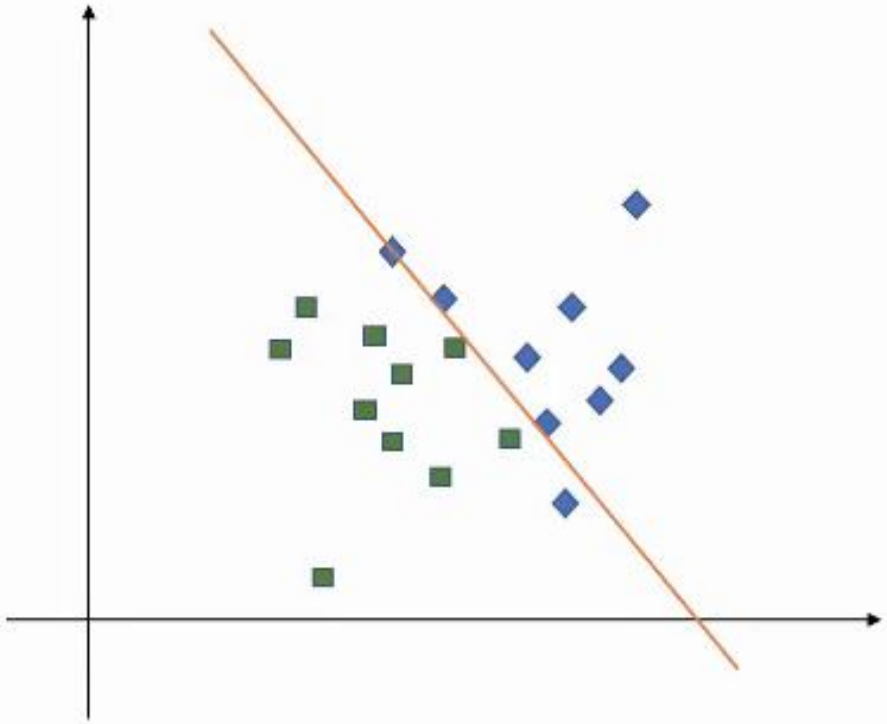
High Gamma



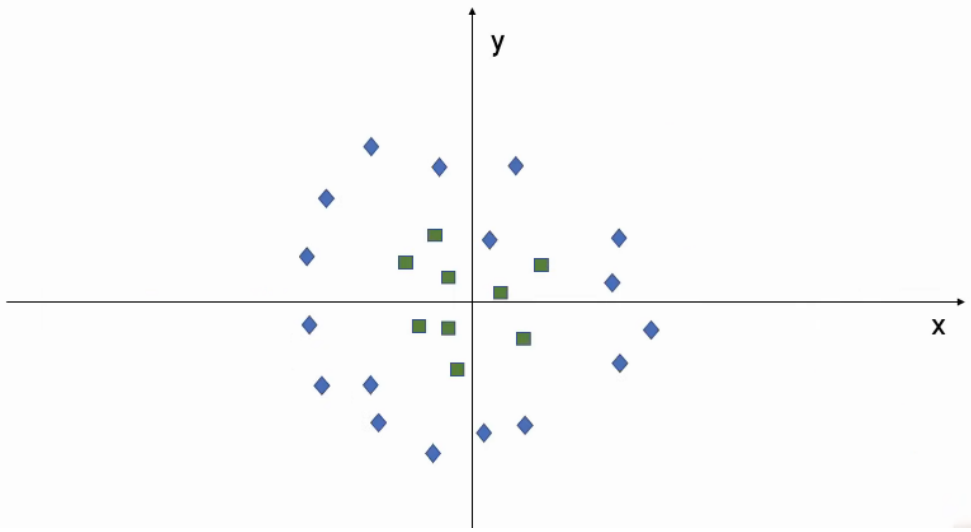
Low Gamma

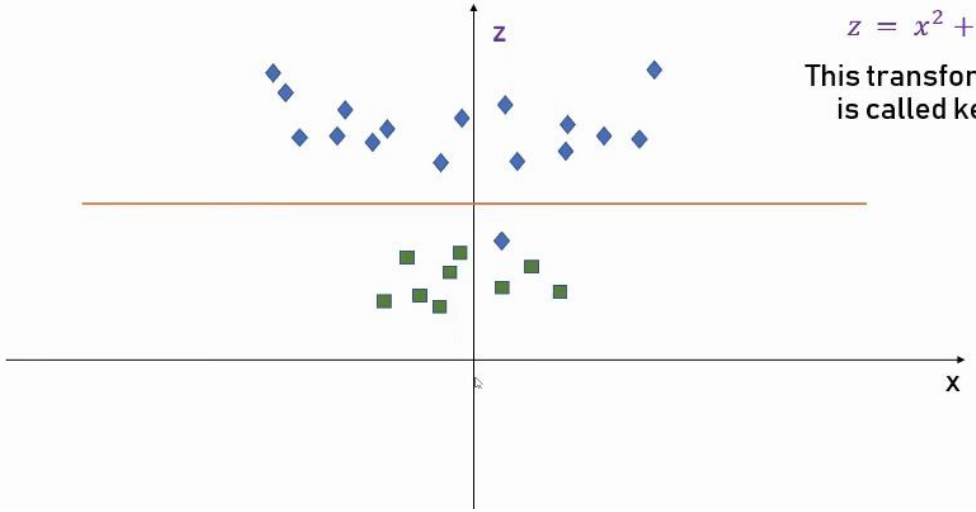


High Regularization (C)



Low Regularization (C)





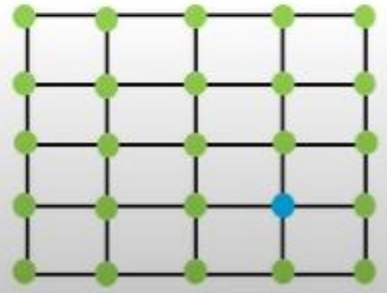
$z = x^2 + y^2$

This transformation is called kernel

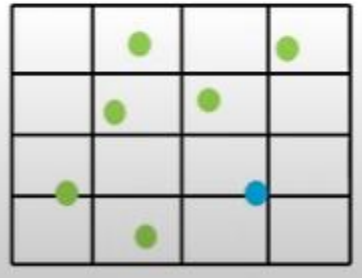
Hyperparameter Tuning

Hyperparameter Tuning refers to the process of choosing the optimum set of hyperparameters for a Machine Learning model. This process is also called **Hyperparameter Optimization.**

Hyperparameter Tuning Types:

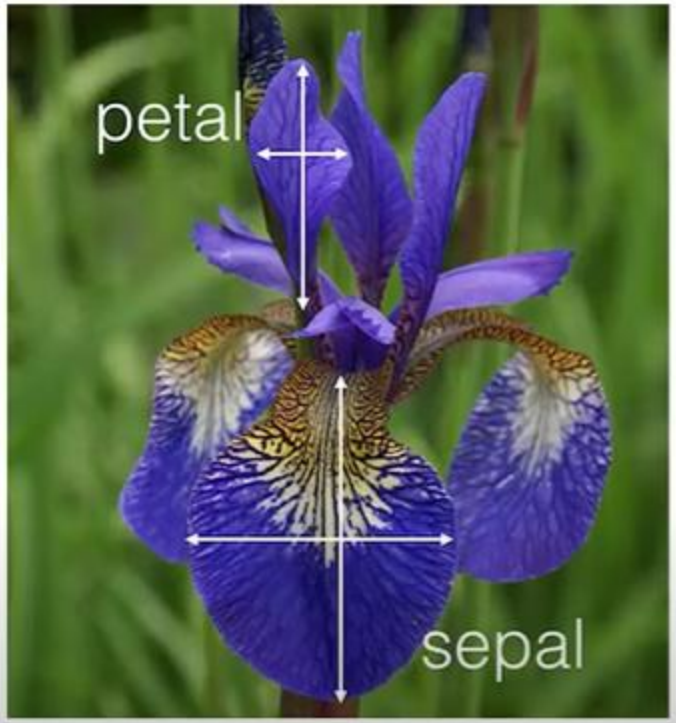


GridSearchCV



RandomizedSeaechCV

sklearn iris flower dataset



features

target label

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	flower
4.6	3.2	1.4	0.2	setosa
5.3	3.7	1.5	0.2	setosa
5.0	3.3	1.4	0.2	setosa
7.0	3.2	4.7	1.4	versicolor
6.4	3.2	4.5	1.5	versicolor

Model selection

Random Forest

SVM

Decision Tree

Logistic Regression

Naïve Bayes

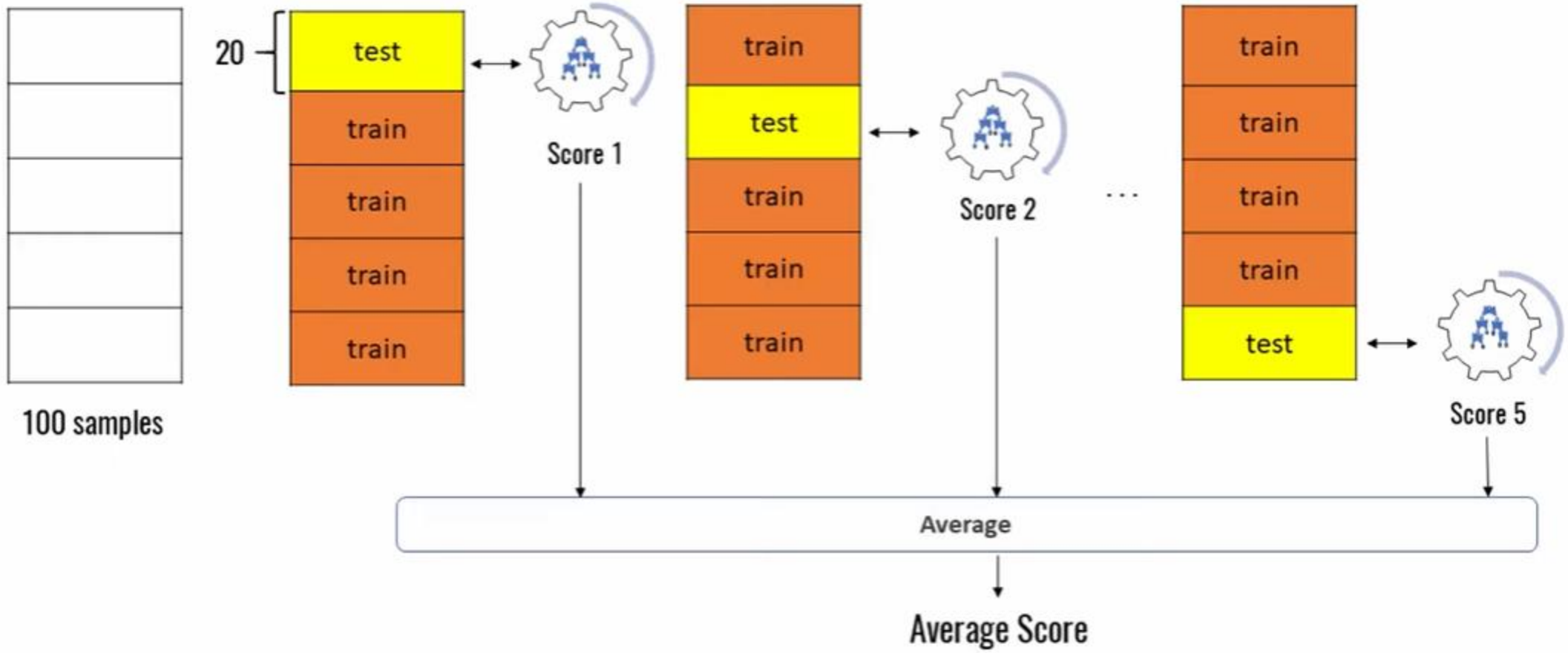
Hyperparamet er Tuning

SVM

example

```
model = svm.SVC(kernel='rbf',C=30,gamma='auto')
```

Parameter	Values
Kernel	'rbf', 'linear', 'poly'
C	Integer
Gamma	float



5 fold cross validation